NATIONAL GRID ANALYTICS | SPECIAL FESTIVE STUDY

The Diwali Electricity Paradox

Why does India's national power consumption plummet by 15% to 19% during the Festival of Lights?

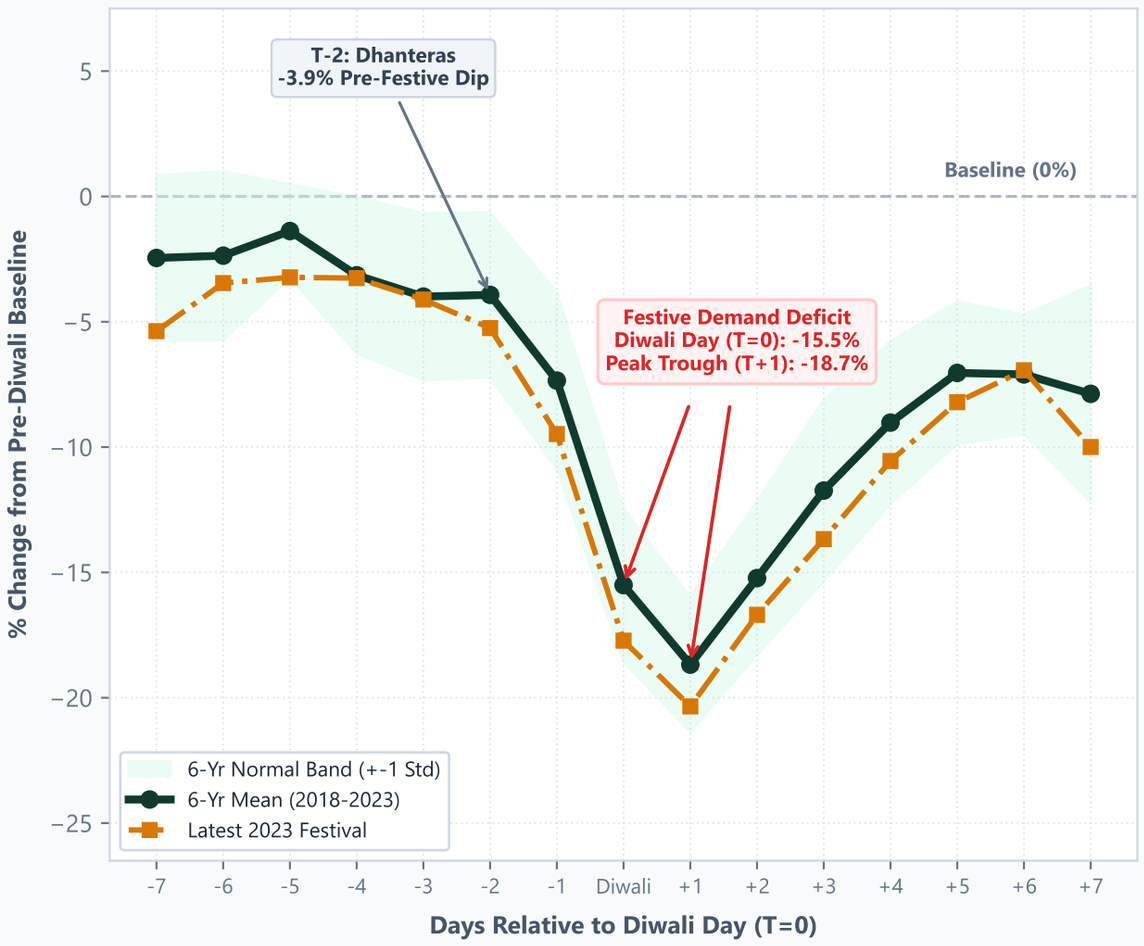
-15.5%
Diwali Day Deficit
11-yr avg drop on festival day

-18.7%
Trough at T+1
Govardhan Puja peak national dip

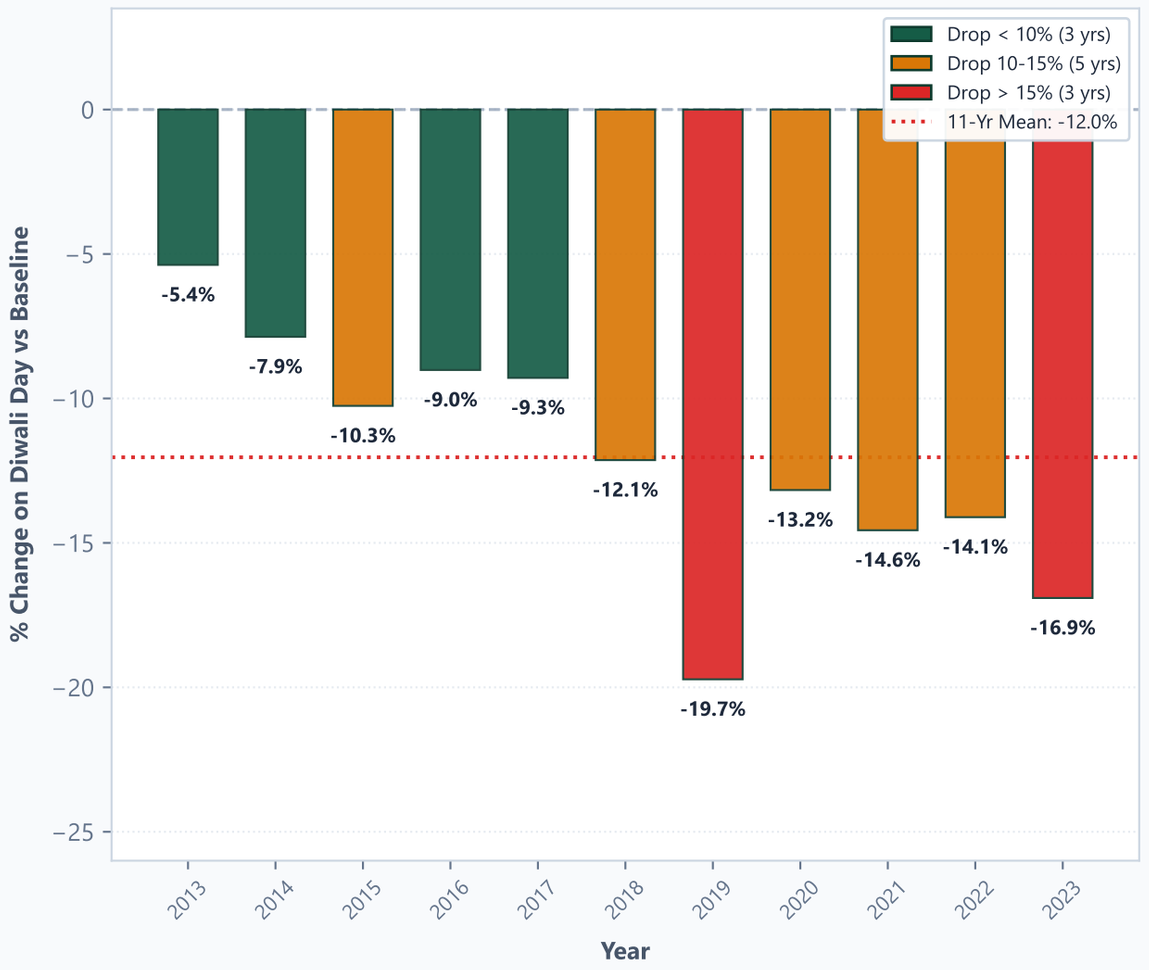
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Decadal Consistency
Every year 2013-2023 saw a drop

-47.9%
Peak State Dip
Daman & Diu (TN -32%, GJ -29%)

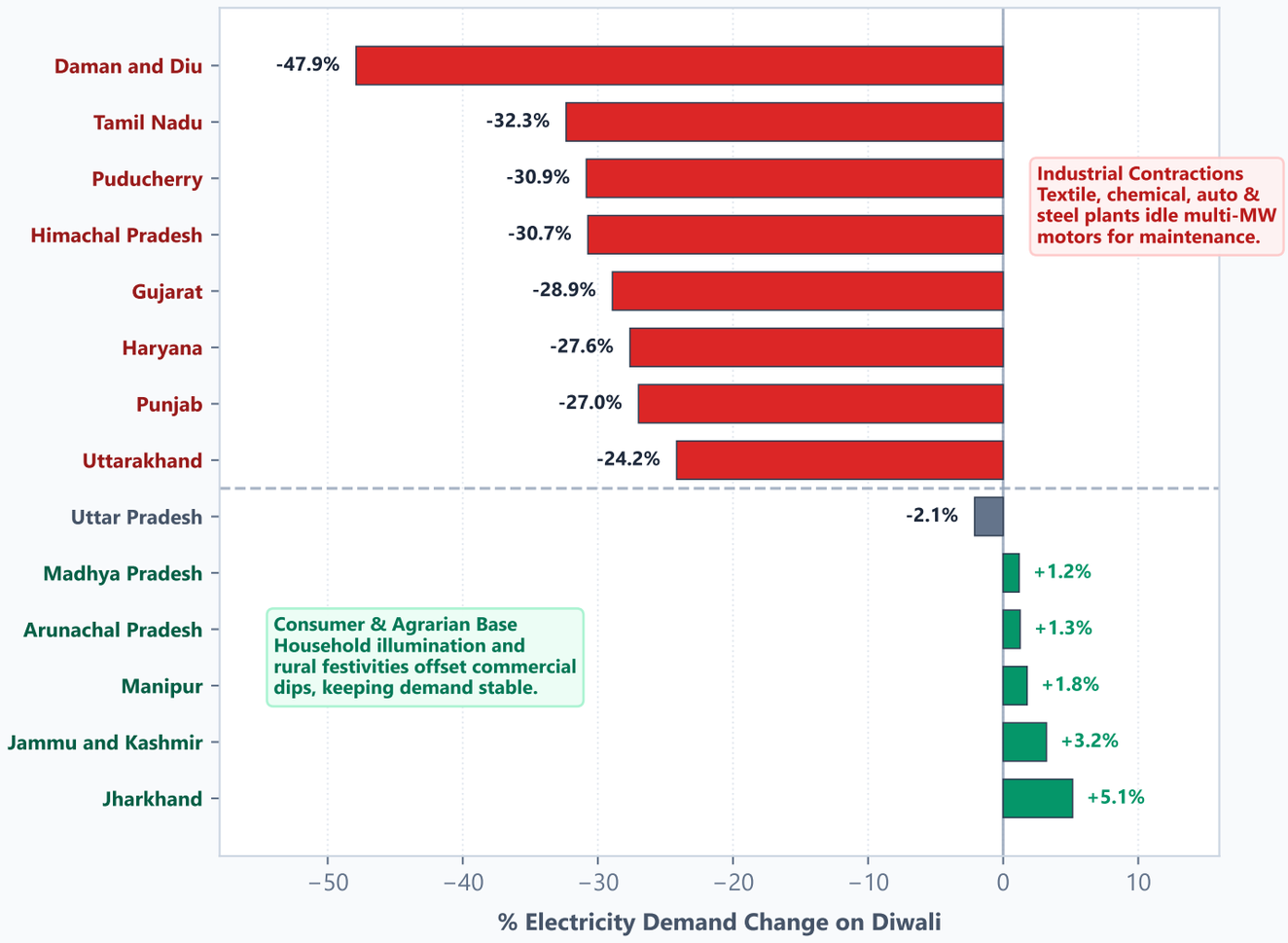
A. The 15-Day Demand Trajectory (T-7 to T+7)



B. 11-Year Historical Consistency (2013-2023)



C. State Divergence: Industrial Halts vs Consumer Resilience



D. Grid Physics: The 3 Drivers of the Paradox

1. Commercial & Industrial (C&I) Shutdown Effect

IMPACT: ~686 MU/day Industrial Deficit

Commercial and industrial customers account for ~49% of all electricity consumed in India. Continuous-process plants (textiles, chemicals, auto manufacturing, steel rolling) in Gujarat, Tamil Nadu, and Maharashtra initiate planned annual maintenance halts from Dhanteras through Bhai Dooj. The idling of multi-megawatt industrial motors creates a massive deficit that domestic lighting cannot replace.

2. The Modern LED Efficiency Revolution

IMPACT: Barely ~6 MU/day Lighting Surge (<1% of deficit)

While public intuition assumes decorative lighting creates massive power surges, modern LED serial strings draw only 3W to 10W per 10-meter strip (vs 60W-100W for legacy incandescents). Even if 100 million households power 3 LED strings simultaneously, the total national peak load is ~1,500 MW for 4 hours (~6 MU), which is barely 1% of the ~686 MU daily industrial drop.

3. Agricultural Pumping Holidays

IMPACT: ~120-180 MU/day Agricultural Load Reduction

Agricultural irrigation pumpsets represent ~18% to 22% of total load in major states (Punjab, Haryana, Rajasthan, Madhya Pradesh). Tubewells and rural feeder pumps are largely switched off by farming communities during festival celebrations, further amplifying the grid-wide demand suppression.

EXECUTIVE GRID SUMMARY & UTILITY IMPLICATIONS

High-Voltage & Frequency Control: For Grid-India and SLDCs, Diwali is not a scarcity problem, but an over-frequency and voltage control challenge.
Ferranti Effect on EHV Lines: With 15% to 19% of national load vanishing, transmission networks face severe capacitive voltage surges from lightly loaded lines.
Systemic Back-down: Grid operators back down thermal baseload, ramp down hydro, and switch in 765 kV / 400 kV bus reactors to maintain 50.0 Hz stability.

Data: Grid-India / NLDC Open Data (3,955 Days, 132,600+ State Records) | Analysis: Analysis Hub | Author: Yash Vasudeva